

Kanu Madhok

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PROFESSIONAL EXPERIENCE

Walmart Data Ventures

Chicago, IL · Sep 2024 – Present

Senior Data Analyst

- Replaced ~20 distinct recruitment workflows I had been running by hand 1–2× weekly with end-to-end automation on a 68-table BigQuery platform. Stratified-sampling pipelines recruited 280,000+ panelists across 29 categories with zero duplicates. A recurring 3-hour task now takes ~3 minutes of setup.
- Designed and shipped to production a self-service analytics agent over 67 BigQuery tables with human-in-the-loop validation, 8 deterministic SQL rules, and a 23-case golden-query evaluation suite. Partnered with Product, Data Science, and business stakeholders to translate ambiguous business questions into agent-ready specs; approved by Sr. Director and product leadership and in active use by business teams.
- Evolved the analytics skill into a registry-governed semantic layer over BigQuery: YAML definitions plus SQLite vocabulary snapshots define legal dimensions, values, and table wiring; a single LLM call maps the question to strict JSON intent (the model never writes SQL); a deterministic gate validates and canonicalizes intent before a deterministic compiler emits the SQL — same question, same answer, every time. New or changed definitions pass a human review gate before entering the registry.
- Built and deployed to production a headless KPI monitor on a 6-hour cron that detects threshold breaches, runs root-cause analysis, and posts Slack alerts; end-to-end observability via decision logging and BigQuery state persistence, so the agent has no LLM-memory reliance.
- Built and shipped to production an autonomous Jira ticket-resolution agent with retrieval (FAISS + BM25) over 11,000 historical SQL queries; closed 400+ data requests across 8 categories over 15 months, cutting routine turnaround from 30–60 minutes to under 10 minutes each.

University of Chicago — Data Science Institute

Chicago, IL · Jan – Sep 2024

Graduate Researcher, Data & Democracy Initiative

- Persona-driven multi-LLM (OpenAI + Gemini) chatbot for a 500+ participant donation experiment; controlled A/B tests persuaded 15% to donate to an opposing cause.

Innovare (EdTech Startup)

Chicago, IL · Oct 2022 – Mar 2023

Data Scientist

- Logistic-regression risk model + Looker dashboards for 10+ districts (5,000+ students); retention +7%, failures –15%.

SELECTED PROJECT

- Live deployed text-to-SQL copilot (Streamlit + Gemini 2.5 + RAG over historical queries, BigQuery execution, auto-visualization): live demo — <https://sql-rag-frontend-simple-481433773942.us-central1.run.app/>

SKILLS

Data: Python, SQL, BigQuery, PySpark, dbt, Airflow, Databricks, MySQL

Evaluation & data quality: evaluation suites (golden sets, human-in-the-loop validation), decision logging / observability, context engineering, RAG (FAISS + BM25 + RRF)

LLMs: Claude (Opus / Sonnet / Haiku via Wibey), Gemini (2.5 Pro / Flash via Vertex AI), OpenAI

AI-assisted development: Claude Code, Cursor

APIs / triggers / glue: FastAPI, Cloud Run, BigQuery scheduled queries, Slack webhooks, cron, Playwright SSO, JSONL state stores, CI/CD (pytest, version control)

Viz / reporting: Streamlit, Power BI (DAX, XMLA TMSL), Tableau, LookML, Matplotlib, React Recharts

EDUCATION

The University of Chicago — M.S., Applied Data Science (GPA 4.0)

Chicago, IL

Loyola University Chicago — B.S., Information Systems & Economics (GPA 3.5)

Chicago, IL